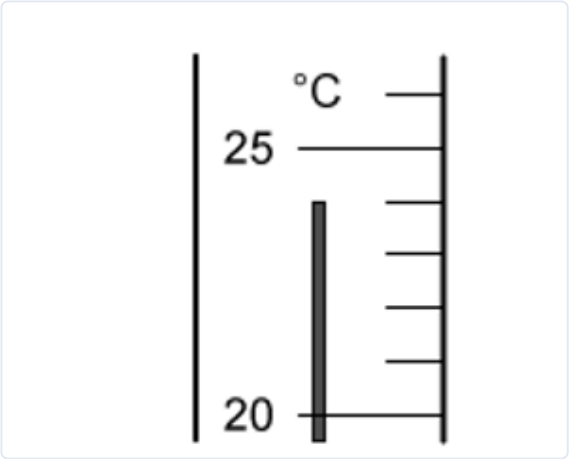
CIE AS PHYSICS (9702) • QUESTION PAPER

1.3 Errors & Uncertainties

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Multiple Choice (Paper 1) **Total:** 22 Questions (22 Marks)

Question 1 Easy [1 mark]

The diagram shows part of a thermometer.



What is the correct reading on the thermometer and the uncertainty in this reading?

	Reading/°C	Uncertainty in reading/°C
A	24	±1
B	24	±0.1
C	24.0	±0.2
D	24.0	±0.5

A

B

C

D

Question 2 Easy [1 mark]

In an experiment to determine the acceleration of free fall using a falling body, what would lead to a value that is too large?

A air resistance

B dimensions of the body are too large

C measured distance longer than true distance

D measured time longer than true-time

Question 3 Easy [1 mark]

A metre rule is used to measure the length of a piece of wire. It is found to be 70 cm long to the nearest millimetre.

How should this result be recorded in a table of results?

A 0.7 m

B 0.70 m

C 0.700 m

D 0.7000 m

Question 4 Easy [1 mark]

An experiment is done to measure the acceleration of free fall of a body from rest.

Which measurements are needed?

A The height of fall and the time of fall

B The height of fall and the weight of the body

C The mass of the body and the height of fall

D The mass of the body and the time of fall

Question 5 Easy [1 mark]

A micrometer is used to measure the diameters of two cylinders

diameter of first cylinder = 12.78 ± 0.02 mm

diameter of second cylinder = 16.24 ± 0.03 mm

The difference in the diameters is calculated.

What is the uncertainty in this difference?

A ± 0.01 mm

B ± 0.02 mm

C ± 0.03 mm

D ± 0.05 mm

Question 6 Easy [1 mark]

A student carries out a series of determinations of the acceleration of free fall g. The table shows the results.

g / m s^{-2}
4.91
4.89
4.88
4.90
4.93
4.92

What can be said about this experiment?

A it is accurate and precise

B it is accurate but not precise

C it is not accurate and not precise

D it is not accurate but is precise

Question 7 Medium [1 mark]

The measurement of a physical quantity may be subject to random errors and systematic errors.

Which statement is correct?

A random errors can be reduced by taking the average of several measurements

B random errors are always caused by the person taking the measurement

C a systematic error cannot be reduced by adjusting the apparatus

D a systematic error results in a different reading each time the measurement is taken

Question 8 Medium [1 mark]

In an experiment, a radio-controlled car takes 2.50 ± 0.05 s to travel 40.0 ± 0.1 m.

What is the car's average speed and the uncertainty in this value?

A $16 \pm 1\ \text{m s}^{-1}$

B $16.0 \pm 0.2\ \text{m s}^{-1}$

C $16.0 \pm 0.4\ \text{m s}^{-1}$

D $16.0 \pm 0.36\ \text{m s}^{-1}$

Question 9 Medium [1 mark]

Four students each made a series of measurements of the acceleration of free fall *g*. The table shows the results obtained.

Which set of results could be described as precise but not accurate?

	g / ms^{-2}			
A	9.81	9.79	9.84	9.83
B	9.81	10.12	9.89	9.94
C	9.45	9.21	8.99	8.76
D	8.45	8.46	8.50	8.41

A

B

C

D

Question 10 Medium [1 mark]

Which experimental technique reduces the systematic error of the quantity being investigated?

A adjusting an ammeter to remove its zero error before measuring a current

B measuring several internodal distances on a standing wave to find the mean internodal distance

C measuring the diameter of a wire repeatedly and calculating the average

D timing a large number of oscillations to find a period

Question 11 Medium [1 mark]

A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

length / mm	892, 891, 892, 891, 891, 892

Are the readings accurate and precise to within 1 mm?

	Results are accurate to within 1 mm	**Results are precise to within 1 mm**
A	no	no
B	no	yes
C	yes	no
D	yes	yes

A

B

C

D

Question 12 Medium [1 mark]

The Young modulus of the material of a wire is to be found. The Young modulus E is given by the equation below.

$$E = \frac{4Fl}{\pi d^2 x}$$

The wire is extended by a known force and the following measurements are made.

Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	Measurement	Symbol	Value
A	length of wire before force applied	l	2.043 ± 0.002 m
B	diameter of wire	d	0.54 ± 0.02 mm
C	force applied	F	19.62 ± 0.01 N
D	extension of wire with force applied	x	5.2 ± 0.2 mm

A

B

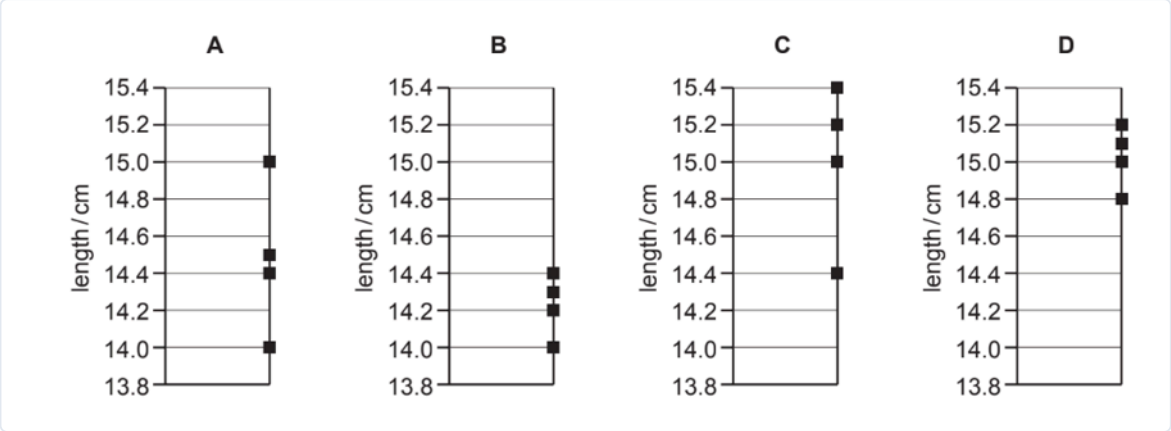
C

D

Question 13 Medium [1 mark]

Four different students use a ruler to measure the length of a 15.0 cm pencil. Their measurements are recorded on four different charts.

Which chart shows measurements that are precise but **not** accurate?



A

B

C

D

Question 14 Medium [1 mark]

What is the ratio $\frac{10^2\ \text{THz}}{10^{-4}\ \text{pHz}}$?

A 10^{-30}

B 10^{-2}

C 10^{2}

D 10^{30}

Question 15 Hard [1 mark]

A student uses a digital ammeter to measure a current. The reading of the ammeter is found to fluctuate between 1.98 A and 2.02 A.

The manufacturer of the ammeter states that any reading has a systematic uncertainty of ±1%.

Which value of current should be quoted by the student?

A (2.00 ± 0.01) A

B (2.00 ± 0.02) A

C (2.00 ± 0.03) A

D (2.00 ± 0.04) A

Question 16 Hard [1 mark]

The uncertainty in the value of the momentum of a trolley passing between two points X and Y varies with the choice of measuring devices.

Measurements for the same trolley made by different instruments were recorded.

1	distance between X and Y using a metre rule with cm divisions = 0.55 m
2	distance between X and Y using a metre rule with mm divisions = 0.547 m
3	timings using a wristwatch measuring to the nearest 0.5 s at X = 0.0 s and at Y = 4.5 s
4	timings using light gates measuring to the nearest 0.1 s at X = 0.0s and at Y = 4.3 s
5	mass of trolley using a balance measuring to the nearest g = 6.4×10^{-2} kg
6	mass of trolley using a balance measuring to the nearest 10g = 6×10^{-2} kg

Which measurements, one for each quantity measured, leads to the least uncertainty in the value of the momentum of the trolley?

A 1, 3 and 6

B 1, 4 and 6

C 2, 3 and 6

D 2, 4 and 5

Question 17 Hard [1 mark]

In an experiment to determine the acceleration of free fall *g*, a ball bearing is held by an electromagnet. When the current to the electromagnet is switched off, a clock starts and the ball bearing falls. After falling a distance *h*, the ball bearing strikes a switch to stop the clock which measures the time *t* of the fall.

If systematic errors cause *t* and *h* to be measured incorrectly, which error **must** cause *g* to appear greater than 9.81 m s^{-2}?

A *h* measured as being **smaller** than it actually is and *t* is measured correctly

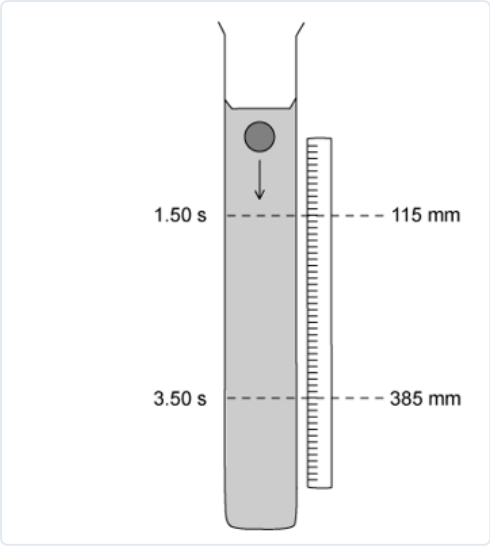
B *h* measured as being **smaller** than it actually is and *t* measured as being **larger** than it actually is

C *h* measured as being **larger** than it actually is and *t* measured as being **larger** than it actually is

D *h* is measured correctly and *t* measured as being **smaller** than it actually is

Question 18 Hard [1 mark]

The diagram shows an experiment to measure the speed of a small ball falling at constant speed through a clear liquid in a glass tube.



There are two marks on the tube. The top mark is positioned at 115 ± 1 mm on the adjacent rule and the lower mark at 385 ± 1 mm. The ball passes the top mark at 1.50 ± 0.02 s and passes the lower mark at 3.50 ± 0.02 s.

The constant speed of the ball is calculated to be 135 mm s^{-1}

Which expression calculates the fractional uncertainty in the value of this speed?

A $\frac{2}{270} + \frac{0.04}{2.00}$

B $\frac{1}{270} + \frac{0.02}{2.00}$

C $\frac{1}{115} + \frac{0.02}{1.50}$

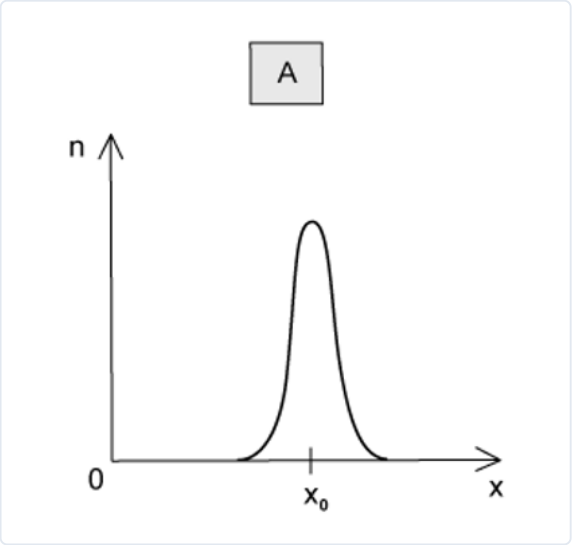
D $\frac{1}{385} + \frac{0.02}{3.50}$

Question 19 Hard [1 mark]

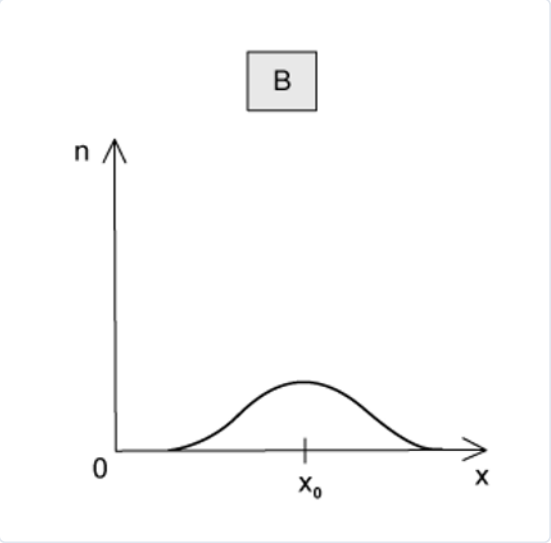
A fixed quantity x_0 is measured many times in an experiment that has experimental uncertainty. A graph is plotted to show the number n of times that a particular value x is obtained.

Which graph could be obtained if the measurement of x_0 has a large systematic error but a small random error?

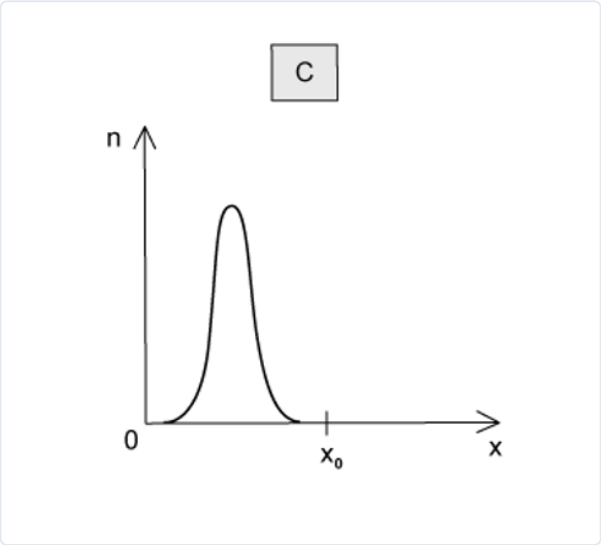
A



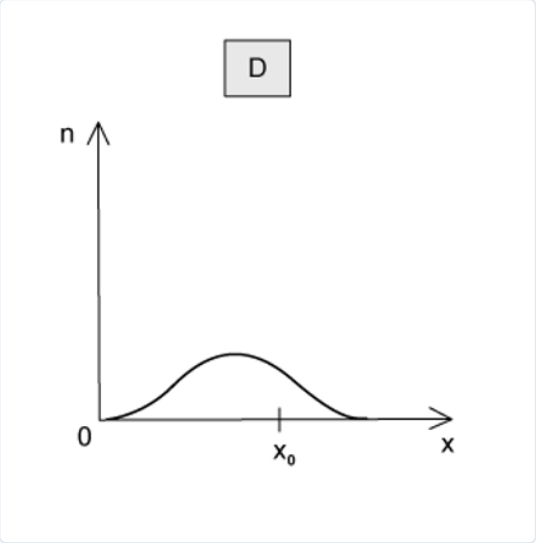
B



C



D



Question 20 Hard [1 mark]

A student finds the density of a liquid by measuring its mass and volume. The following is a summary of his measurements.

mass of empty beaker = (20 ± 1) g

mass of beaker + liquid = (70 ± 1) g

volume of liquid = (10.0 ± 0.5) cm^3

He correctly calculates the density of the liquid as 5.0 g cm^{-3}.

What is the uncertainty in this value?

A 0.3 g cm^{-3}

B 0.5 g cm^{-3}

C 0.6 g cm^{-3}

D 2.6 g cm^{-3}

Question 21 Hard [1 mark]

A steel wire is stretched in an experiment to determine the Young modulus for steel.

The uncertainties in the measurements are given below.

Measurement	Uncertainty
load on wire	±2%
length of wire	±0.2%
diameter of wire	±1.5%
extension	±1%

What is the percentage uncertainty in the Young modulus?

A 1.3%

B 1.8%

C 4.7%

D 6.2%

Question 22 Hard [1 mark]

A mass is dropped from rest and falls through a distance of 2.0 m in a vacuum. An observer records the time taken for the mass to fall through this distance using a manually operated stopwatch and repeats the measurements a further two times. The average result of these measured times, displayed in the table below, was used to determine a value for the acceleration of free fall. This was calculated to be 9.8 m s^{-2}.

	First measurement	Second measurement	Third measurement	Average
Time/ s	0.6	0.73	0.59	0.64

Which statement best relates to the experiment?

A The measurements are precise and accurate with no evidence of random errors.

B The measurements are not accurate and not always recorded to the degree of precision of the measuring device but the calculated experimental result is accurate.

C The measurements are not always recorded to the degree of precision of the measuring device but are accurate. Systematic errors may be present.

D The range of results shows that there were random errors made but the calculated value is correct so the experiment was successful.